

5. Specification

Any technical spec shall refer to Nordic's official documents as final reference. Contents below are from "[nRF52840 Product Specification v1.7](#)", please click to download full spec.

5.1. Absolute Maximum Ratings

	Note	Min.	Max.	Unit
Supply voltages				
VDD		-0.3	+3.9	V
VDDH		-0.3	+5.8	V
VBUS		-0.3	+5.8	V
VSS			0	V
I/O pin voltage				
V _{I/O} , VDD ≤ 3.6 V		-0.3	VDD + 0.3	V
V _{I/O} , VDD > 3.6 V		-0.3	3.9	V
NFC antenna pin current				
I _{NFC1/2}			80	mA
Radio				
RF input level			10	dBm
Environmental aQFN™ 73 package				
Storage temperature		-40	+125	°C
MSL	Moisture Sensitivity Level		2	
ESD HBM	Human Body Model		2	kV
ESD HBM Class	Human Body Model Class		2	
ESD CDM	Charged Device Model		450	V
Flash memory				
Endurance		10 000		write/erase cycles
Retention at 85 °C		10		years

5.2. Operating Conditions

Symbol	Parameter	Min.	Nom.	Max.	Units
VDD	VDD supply voltage, independent of DCDC enable	1.7	3.0	3.6	V
VDD _{POR}	VDD supply voltage needed during power-on reset	1.75			V
VDDH	VDDH supply voltage, independent of DCDC enable	2.5	3.7	5.5	V
VBUS	VBUS USB supply voltage	4.35	5.0	5.5	V
t _{R_VDD}	Supply rise time (0 V to 1.7 V)			60	ms
t _{R_VDDH}	Supply rise time (0 V to 3.7 V)			100	ms
TA	Operating temperature	-40	25	85	°C
T _J	Junction temperature			90	°C

*** The on-chip power-on reset circuitry may not function properly for rise times longer than the specified maximum.